

Master's Comprehensive Examination

Part II

Applied (90 minutes)

December 1, 2007

Instructions: Answer any **three** out of the four questions. Open book and open notes.

Question 1

The weight of a poultry farmer's chicken eggs is normally distributed with a mean weight of 58 grams, and a standard deviation of 12 grams. A feed salesman claims that using their new diet will increase the mean weight of eggs. To see if there is any evidence to support the salesman's claim, the farmer feeds her chickens with the new diet, and then takes a sample of one egg from each of 36 chickens. If the sample mean weight of the 36 eggs is greater than 61 grams, she will feel that this is sufficient evidence for her to conclude that the diet increases the mean weight of eggs. The chicken farmer is using a statistical test at a given α level.

- What is the value of α for this test _____
- If in fact the new feed increases average weight to $\mu = 60$ grams, what is the power of the test? _____
- If she carries out the test and observes a sample mean = 62 grams, what is the p-value of the test _____

Question 2

A medical researcher seeks to compare the efficacy of three treatments T1, T2, T3 for treating a partially blocked artery.

T1: inserts an uncoated stent; T2: inserts a coated stent; T3: No stent.

A random sample of six cardiologists are assigned at random, two cardiologists to each treatment. A random sample of 60 patients are picked and assigned at random, 10 patients to each cardiologist. The data layout is as follows:

	Treatment T1		Treatment T2		Treatment T3	
	Cardiol 1	Cardiol 2	Cardiol 3	Cardiol 4	Cardiol 5	Cardiol 6
Number of Patients	10	10	10	10	10	10

Write out the ANOVA table showing Sources of Variability, Degrees of Freedom, Expected Mean Squares, and show which F-ratios to take.